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# Can Decision Models Understand Stance ?

## Evaluating Jev Against General Purpose LLMs

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# What is Stance Detection ?

Stance detection is the task of identifying a person's position or opinion toward a particular target.

The paper considers three possible stance labels:

- Favor: The text supports or agrees with the target.
- Against: The text opposes or disagrees with the target.
- Neutral: The text does not clearly support or oppose the target.

Example:

Target: Online education is beneficial.

- “It provides flexible learning.” → Favor
- “It reduces direct interaction.” → Against
- “Online education is widely used.” → Neutral

**FAVOR**

Supports or agrees with the target

**AGAINST**

Opposes or criticizes the target

**NEUTRAL**

No clear position towards the target

# Why is Stance Detection Challenging?

-Identifying a person's stance is not always straightforward.

**Context matters**

The meaning of a statement may depend on earlier messages.

**Indirect opinions**

People may express agreement or disagreement without using direct words.

**Different targets**

A statement may support one idea but oppose another.

**Ambiguous statements**

Some text may not clearly express any position.

**CONVERSATION UNDERSTANDING**

Speaker A : “Online education is very useful”

Speaker B : “I completely disagrees”

The speaker and conversation history may be necessary to interpret the actual opinion.

**The earlier message helps to interpret the reply**

# Research Motivation and Objective

**Why did the researchers conduct this study?**

- Traditional stance detection approaches often depend on task-specific labelled datasets.
- General-purpose Large Language Models (LLMs) can also perform stance classification using instructions.
- Stance detection has three predefined output labels: Favor, Against and Neutral.
- The researchers investigate whether Jev can perform this classification effectively compared with general-purpose LLMs.

# Related Work : Stance Detection

1. **Target-specific stance detection**
   - Identifies the relationship between a text and a particular target.
   - Example: Understanding whether a text supports or opposes a given topic.
2. **Cross-target stance detection**
   - Transfers knowledge between related targets.
   - Reduces the need for labelled data for every target.
3. **Zero-shot stance detection**
   - Identifies stance toward targets that were not seen during training.
   - Example: VAST dataset.
4. **LLM-based approaches**
   - Use Large Language Models to identify relationships between text and target.
   - Some approaches use LLM-generated knowledge to support stance prediction.

# Related Work : Conversational Stance Detection

## What is Conversational Stance Detection (CSD)?

Conversational stance detection identifies a speaker's position toward a target by considering the conversation and interactions between speakers.

Important research works:

- SRQ: Studies stance in replies and quoted tweets.
- MT-CSD: Models dependencies across multiple conversation turns.
- GLAN: Combines global and local conversational information.
- ZS-CSD: Introduces a Chinese benchmark for stance detection toward unseen targets.
- SITPCL: Combines speaker interaction with target-aware contrastive learning.

# WorkFlow

## DATASETS

VAST
(Individual text)

ZS-CSD
(Conversational)

## INPUT PREPARATION

- Target
- Focus Text
- Conversation History
- Target Type

## Jev-Choice Based Decision

**Task Instruction**

Determine the stance towards the target

**Three Choices**

**FAVOR**
Supports or agrees with the target

**AGAINST**
Opposes or criticizes the target

**NEUTRAL**
No clear position towards the target

**Jev**
Select One Option

## Prediction Extraction

Extract & Validate Selected Label

- FAVOR
- AGAINST
- NEUTRAL

## Evaluation Metrics

Metrics

- F1 Score
- Accuracy

# Datasets , Models and Evalution Metrics

1. **Datasets**
   - VAST: Individual English text – 3,006 test records.
   - ZS-CSD: Chinese conversational text – 2,584 test records; focuses on unseen-target stance detection.
2. **Models**
   - Main Model: Jev
   - General-Purpose LLMs:
     DeepSeek V4.1 Flash, Qwen3.8-Flash, GPT-5.6, Claude Sonnet 5.
   - Fine-Tuned Reference Models: Qwen2.5-3B (LoRA), Qwen2.5-7B (LoRA).
3. **Evaluation Metrics**
   - **Accuracy:** Measures the overall percentage of correct predictions.
   - **Macro-F1 Score:** Measures performance across Favor, Against, and Neutral classes.
   - **Token Usage:** Measures the number of tokens consumed.
   - **API Cost:** Measures the cost of processing the dataset.

# Different Stages

1

## TASK DEFINITION



Identify the stance of the focus text towards a given target.

FAVOR

AGAINST

NEUTRAL

2

## STRUCTURED INPUT CONSTRUCTION



- Prepare the input using:
  - Target
  - Focus text
  - Conversation history
  - Target type
- Preserve the original language and conversation order.

3

## JEV CHOICE BASED DECISION

FAVOR

AGAINST

NEUTRAL

- Provide the task instruction and three stance options to Jev.
- Jev selects exactly one option based on the given criteria:
  - Favor – Support or approval
  - Against – Opposition or criticism
  - Neutral – No clear stance

4

## PREDICTION EXTRACTION



- Extract the selected option from Jev's response.
- Validate the selected label.
- Use it as the final stance prediction.

# Complete Example : Input To Prediction

1. **Target**

   Online education is better than classroom education

2. **Conversation History**

   Speaker A : “Online education gives students more flexibility.”

3. **Focus Text**

   Speaker B : “I completely disagree. Students learn better through direct classroom interaction.”



**Predicted**

**AGAINST**

# Results

| Dataset | Model / setting | Accuracy | Macro-F1 | Input tokens | Output tokens | Cost |
|---|---|---|---|---|---|---|
| **VAST** | Jev | 78.14 | 77.92 | 1,763,143 | 114,228 | $0.0741 |
| | DeepSeek V4.1 Flash | 74.82 | 73.99 | 865,478 | 18,708 | $0.2133 |
| | Qwen3.8-Flash | 74.52 | 73.79 | 763,609 | 28,874 | $0.1021 |
| | GPT-5.6 | **78.51** | **78.08** | 1,178,659 | 33,439 | $3.9010 |
| | Claude Sonnet 5 | 74.82 | 74.51 | 2,279,907 | 251,109 | $4.7051 |
| | Qwen2.5-3B-Instruct (SFT)† | 74.16 | 74.52 | — | — | — |
| | Qwen2.5-7B-Instruct (SFT)† | 79.32 | 79.91 | — | — | — |
| **ZS-CSD** | Jev | 58.40 | 58.43 | 1,846,931 | 98,192 | $0.0776 |
| | DeepSeek V4.1 Flash | 58.28 | 56.89 | 815,813 | 15,920 | $0.1829 |
| | Qwen3.8-Flash | 54.61 | 53.16 | 707,481 | 27,236 | $0.0948 |
| | GPT-5.6 | **74.65** | **73.62** | 1,140,780 | 35,592 | $3.9230 |
| | Claude Sonnet 5 | 72.52 | 72.42 | 2,402,197 | 325,790 | $5.5582 |
| | Qwen2.5-3B-Instruct (SFT)† | 55.76 | 56.08 | — | — | — |
| | Qwen2.5-7B-Instruct (SFT)† | 60.98 | 60.85 | — | — | — |

Table 1: Performance and resource usage on the full test sets of VAST and ZS-CSD. Bold indicates the best results among prompted models. Dashes denote unavailable measurements. †SFT results are from single runs without per-instance predictions.

# Results

| Dataset | Model | Against | Favor | Neutral |
|---|---|---|---|---|
| VAST | Jev | 77.13 | 72.71 | **83.92** |
| | DeepSeek | 74.15 | 66.83 | 81.00 |
| | Qwen | 73.88 | 65.52 | 81.97 |
| | GPT-5.6 | **78.25** | **73.95** | 82.05 |
| | Claude | 75.35 | 70.26 | 77.91 |
| ZS-CSD | Jev | 56.66 | 59.58 | 59.04 |
| | DeepSeek | 54.60 | 51.18 | **64.89** |
| | Qwen | 55.59 | 42.45 | 61.42 |
| | GPT-5.6 | **79.49** | **82.70** | 58.67 |
| | Claude | 75.78 | 77.50 | 63.99 |

Table 2: Per-class F1 (%). Bold indicates the best result for each class within each dataset.

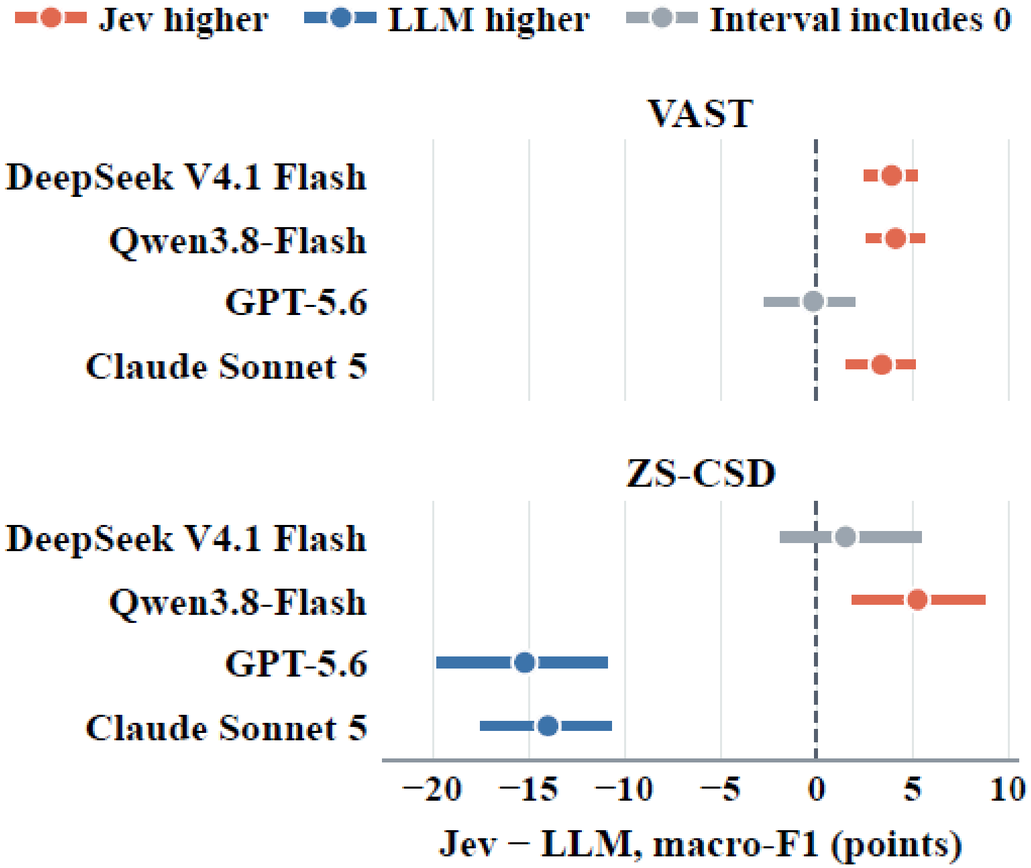



Figure 1: Macro-F1 differences between Jev and each LLM (Jev minus LLM, in percentage points), with paired target-bootstrap 95% intervals.

# Why Does Jev Perform Differently?

1. **Individual Text vs. Conversations**
   - VAST: 77.92% Macro-F1 – competitive performance on individual English text.
   - ZS-CSD: 58.43% Macro-F1 – lower performance on Chinese conversational text.
2. **Difficulty in Identifying Stance Direction**
   - Jev makes 373 Favor–Against classification errors.
   - GPT-5.6 makes 35; Claude Sonnet 5 makes 101.
   - Indicates difficulty interpreting support and opposition between speakers.
3. **Target Complexity**
   - Noun Phrase: 63.49% Macro-F1.
   - Claim: 55.26% Macro-F1.
   - Claims may contain complex semantics, negation and qualifying conditions.
4. **Conversation Length**
   - Performance gap does not consistently increase with conversation length.
   - Reply relationships and stance direction may be more important than the number of turns.

# Real-World Applications & Suitability

1. **Social Media Opinion Monitoring**
   - Classify public opinions towards products, topics or policies.
   - Track Favor, Against and Neutral opinions in incoming posts.
2. **Customer Feedback Analysis**
   - Identify whether customers support or oppose a product feature, service change or company decision.
   - Help businesses monitor customer reactions.
3. **Public Opinion & Policy Analysis**
   - Analyse citizens' positions towards proposed policies, public initiatives or social issues.
   - Monitor changes in public support and opposition.
4. **Online Campaign Monitoring**

-Track audience support or opposition towards campaign messages, proposals and announcements.

# Real-World Applications & Suitability

## More Promising

1. **Individual-text opinion classification**

**77.92** VAST-Macro-F1

**Classifying Favor, Against and Neutral opinions in individual English text, such as social media posts.**

## Relatively Better

2. **Stance toward named entities**

**63.49** ZS-CSD Noun Phrase Macro-F1

**Identifying whether a speaker supports, opposes or is neutral toward a person, organization or product mentioned in a conversation.**

## Limited suitablility

3. **Conversational stance classification**

**58.43** ZS-CSD Overall Macro-F1

**Classifying participants' Favor, Against and Neutral positions in Chinese conversational discussions**

## More Challenging

4. **Complex claim-based stance detection**

**55.26** ZS-CSD Claim Macro-F1

**Identifying opinions toward complex claims, proposals or policy statements.**

# Conclusion

## 1. Limitations

- **Conversational Complexity:** Jev struggles to interpret stance in multi-turn conversations.
- **Stance Direction:** Higher confusion between Favor and Against in ZS-CSD.
- **Target Dependency:** Claim-based targets are more challenging than noun-phrase targets.
- **Cross-Dataset Performance:** Performance varies between individual-text and conversational datasets.

## 2. Key Takeaways

- Jev achieves competitive Macro-F1 on VAST (77.92%).
- Jev outperforms the tested Flash models on both datasets.
- GPT-5.6 performs better on ZS-CSD (73.62%).
- Jev demonstrates substantially lower API cost in the reported experiments.

# Thank you